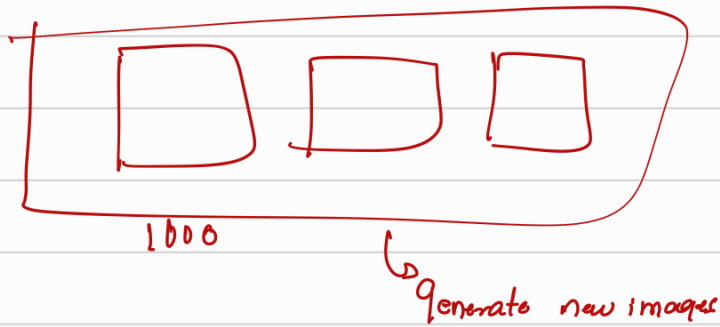
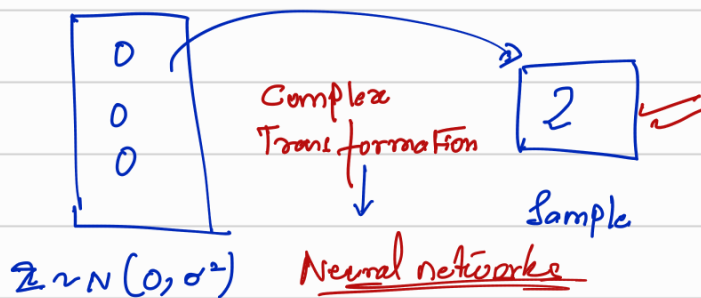


[Generative Adversarial Networks] (GANs)



(n ~~too~~ images) → (generate new images)



Given n number of mnist images, con of learn to generate more images from a simple normal dist.

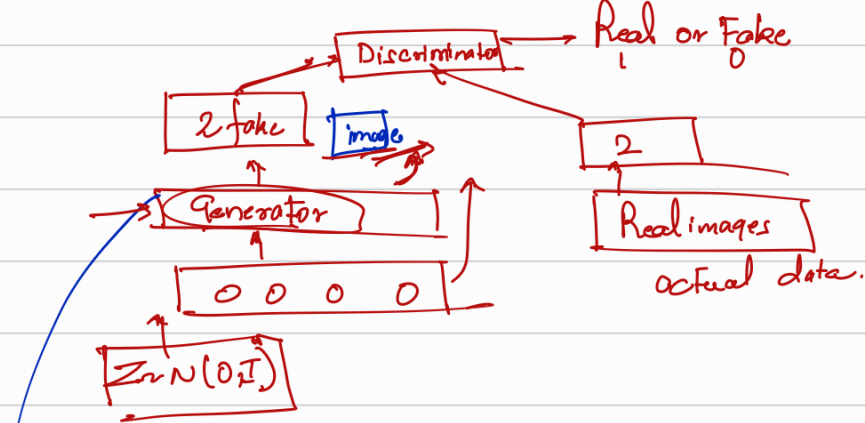


generating the images
from the normal dist

differentiating the image → real or fake.

① generate images such that discriminator finds it hard to discriminate

② to distinguish b/w fake and real in the best possible way.



Objective function of the Generator

Neural network model

image $\rightarrow G \phi(z)$

ϕ NN Parameters
 $D \phi(G \phi(z))$ Parameters

Generator's objective

$$\max_z \log(D \phi(G \phi(z)))$$

$$\max_z p(z) \times \log D \phi(G \phi(z))$$

$$\min_z \int p(z) \times [1 - \log(D \phi(G \phi(z)))]$$

(Discriminator's objective)

$$\max_{x_{data}} E \log(D(x_{data}))$$

Real data

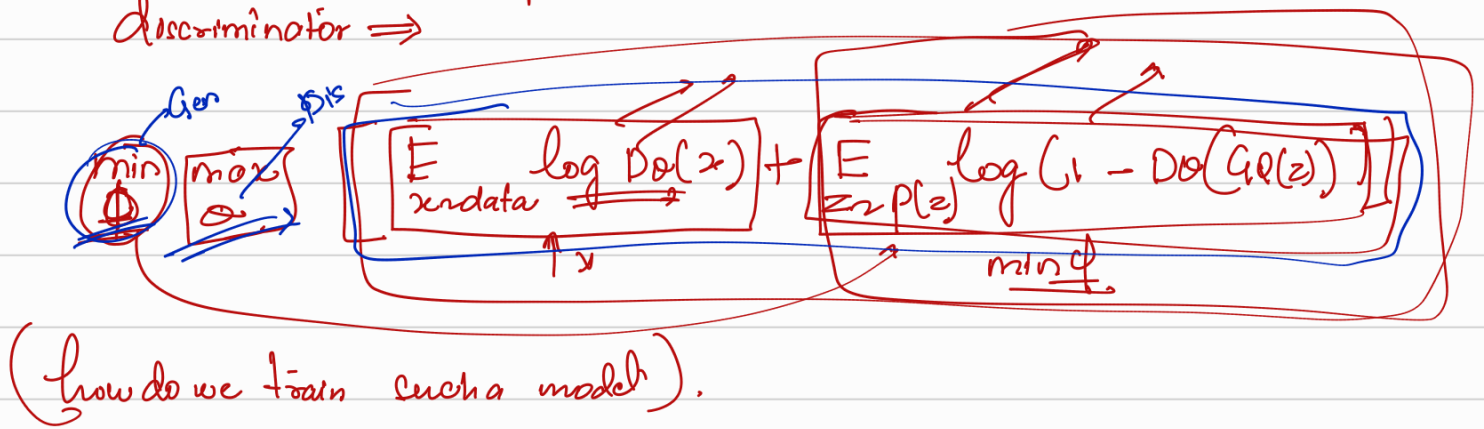
$$\min_z \log D \phi(G \phi(z))$$

$$\max_z E [1 - \log D \phi(G \phi(z))]$$

$$\max_z \left[E_{x_{data}} \log D(x) + E_{z \sim p(z)} \log (1 - D \phi(G \phi(z))) \right]$$

$$\max_z [\log D \phi(G \phi(z))] = \min_z [1 - \log D \phi(G \phi(z))]$$

Combined objective function of the Generator & the discriminator $\Rightarrow$

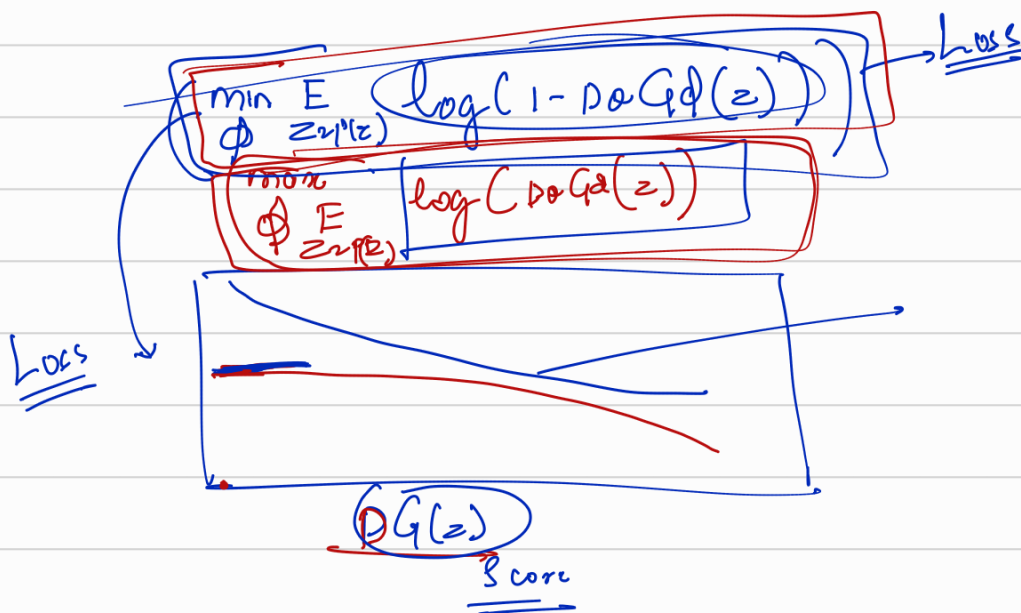


Step 1:

① Gradient ascent on the discriminator:

$$\rightarrow \textcircled{1} \max_{\theta} \left[\mathbb{E}_{x \sim p_{data}} \log D_{\theta}(x) + \mathbb{E}_{z \sim p(z)} \log (1 - D_{\theta}(G_{\phi}(z))) \right]$$

- Gradient descent on the generator.



Procedure for GAN training:

for number of training iterations do:
 for k steps:

① Sample minibatch of m noise samples
 $\{z^1, \dots, z^m\}$.

② Sample minibatch of m real examples
 $\{x^1, \dots, x^m\}$.

③ update the discriminator by ascending the eq:

$$\nabla_{\theta} \mathbb{E}_{\mathbb{P}} \left[\log D_{\theta}(x^i) + \log (1 - D_{\theta} G_{\phi}(z^{i'})) \right]$$

Real
Fake

end for

④ Sample minibatch of m noise samples $\{z^1, \dots, z^m\}$.
 update the generator by Stochastic gradient ascent:

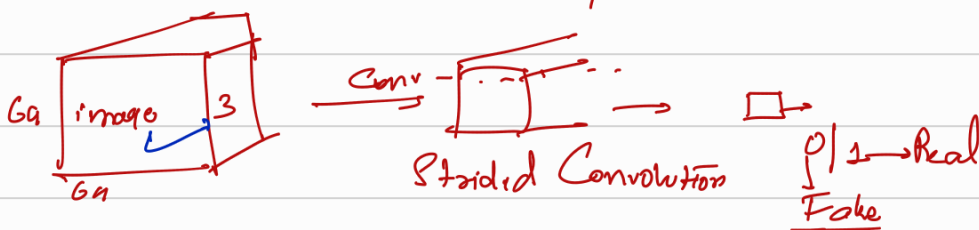
$$\nabla_{\phi} \mathbb{E}_{\mathbb{P}} \left[\log D_{\theta} G_{\phi}(z^i) \right]$$

end for

DGGAN

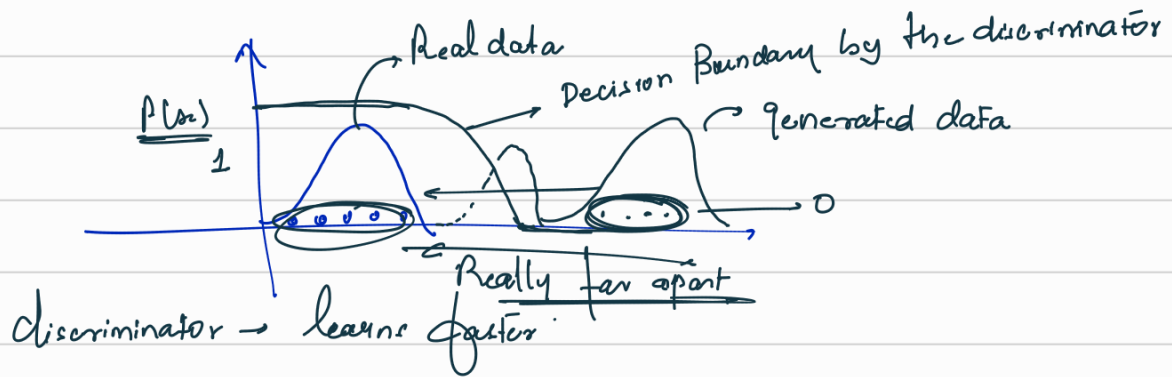
Deep Convolution GANs

Discriminator architecture: (VGGnet / Resnet)

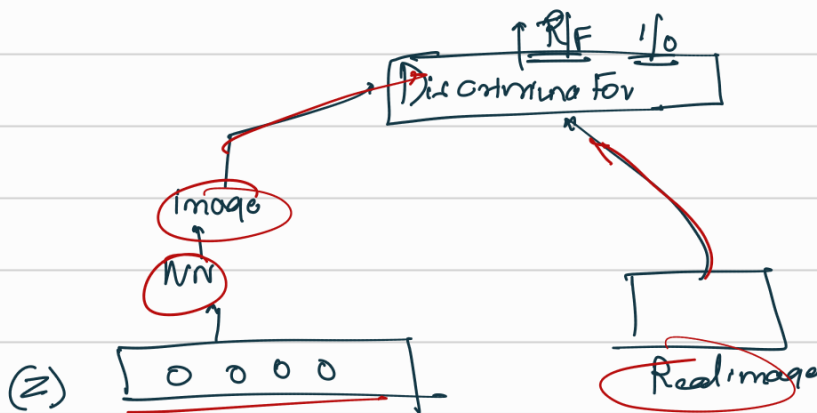


Improved GAN training

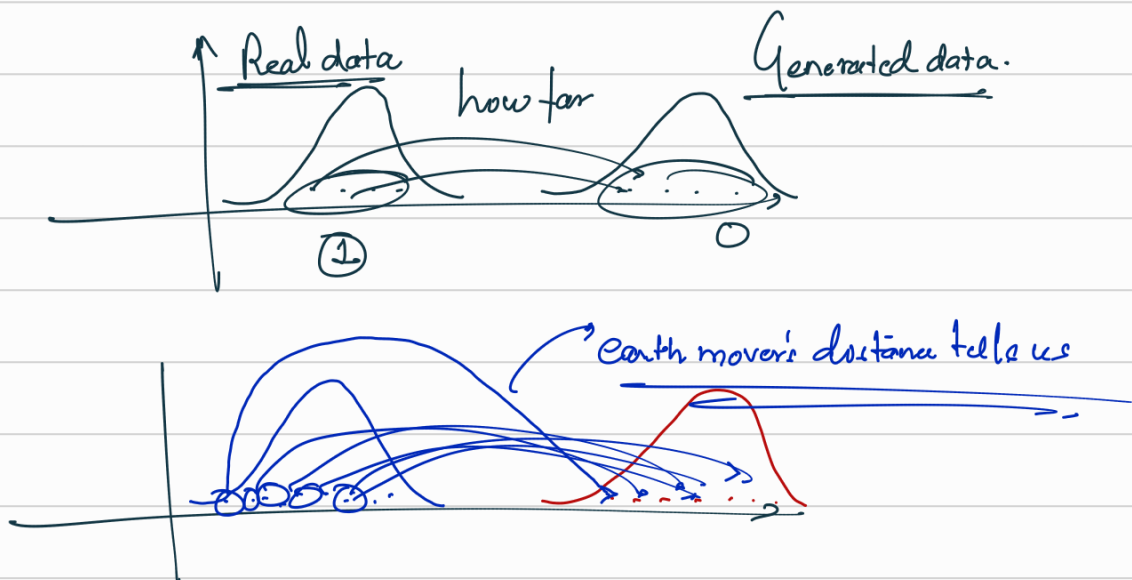
Wasserstein GAN



WGAN



Earth mover's distance



Changes implemented in WGAN compared to normal GAN:

① Discriminator final layer ~~was removed~~ ^{Sigmoid} and instead we used a linear layer.

② Implementation of earth mover's distance

Cool theorem = Kantorovich - Rubinstein Duality.

$$W(p_{data}, p_g) = \sup_{f \text{ Lipschitz constant}} \left[E_{p_{data}}[f(x)] - E_{p_g(x)}[f(x)] \right]$$

1- Lipschitz scalar function

Discriminator neural network

Discriminator Neural network

$$|f(x) - f(y)| \leq |x - y|$$

If I change some data in my real image, the output does not change more than 1 times of the change in the image.

$$y = 2x \rightarrow \text{unit}$$

$$\text{Lipschitz} = 2$$

$$y_1 = 2x_1$$

$$y_2 = 2x_2$$

$$|y_2 - y_1| = 2|x_2 - x_1|$$

$$\frac{1}{n} \sum f(x) - \frac{1}{n} \sum f(x) \left(\frac{\|x_2 - x_1\|_2 - 1}{2} \right) - \frac{1}{n} \sum f(x)$$

the change in the discriminator output value based on a unit change in the image norm

WGAN $\Rightarrow$

Style GANs
Code